

Agentic Discovery of Neural Operator Architectures

AI Scientific Community swarm - 16 virtual labs, citation-based economy of influence, PSO coordinator - paper-grade run, seed 42, RTX 4080

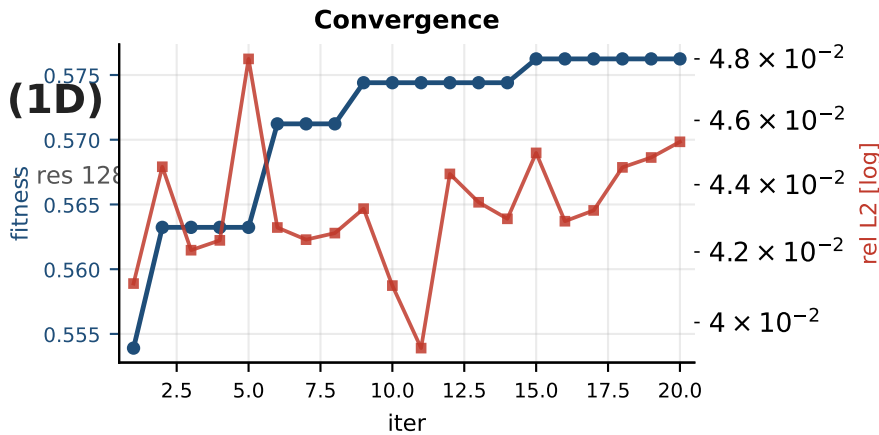
L. F. Loo, U. Braga-Neto - Texas A&M University

Piecewise Regression (1D)

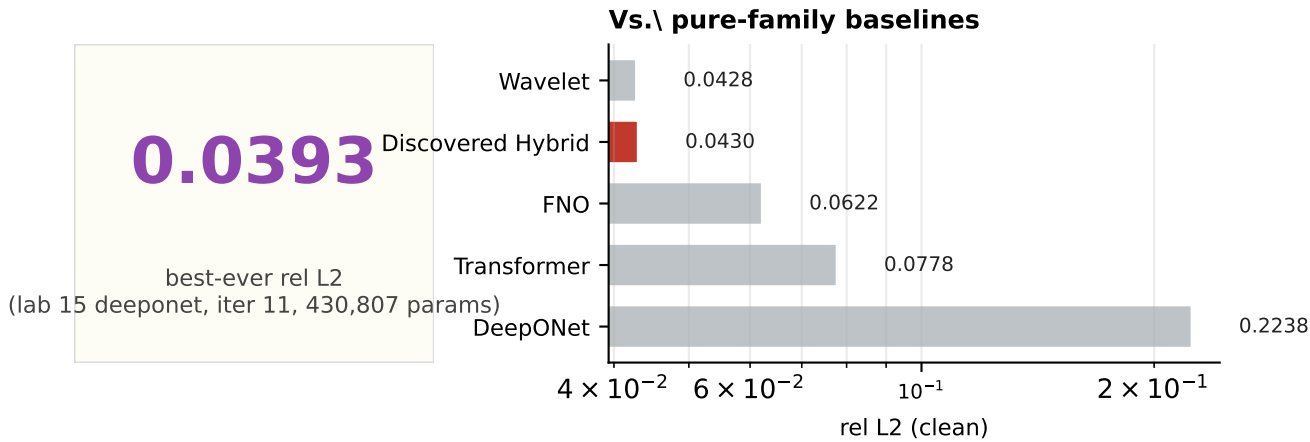
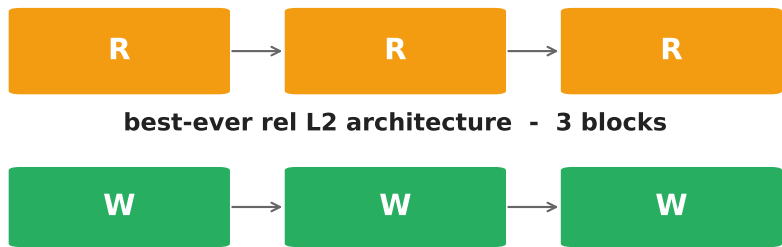
16 labs × 20 iter × 40 epochs × 256 train

Composite fitness: 0.5762

hidden ch.\,= 79, modes = 8, act = relu



Discovered architecture
composite winner - 3 blocks

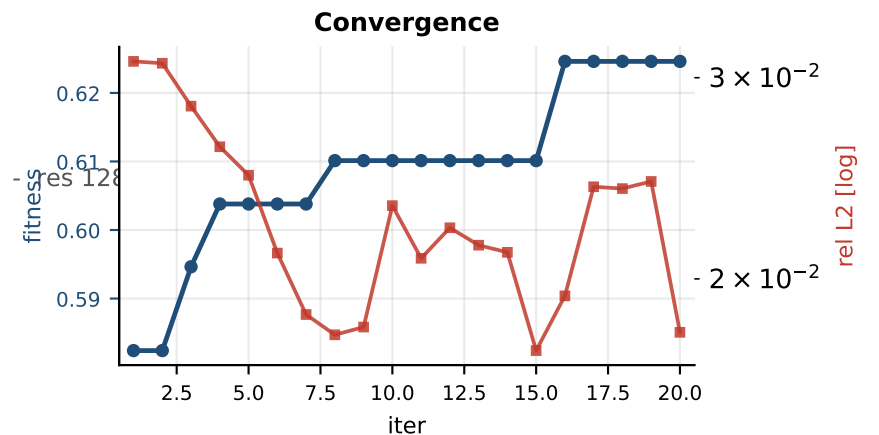


Linear Advection (1D)

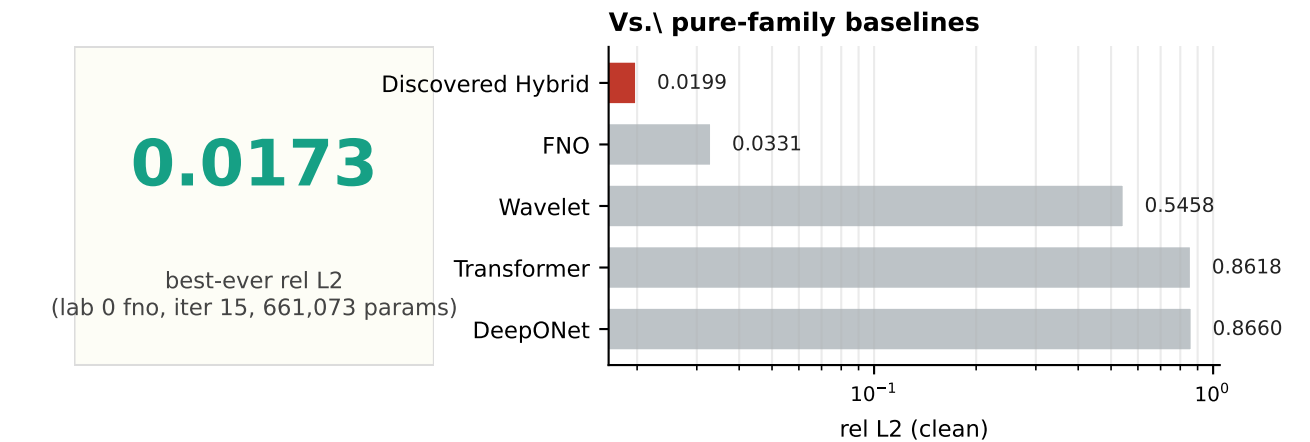
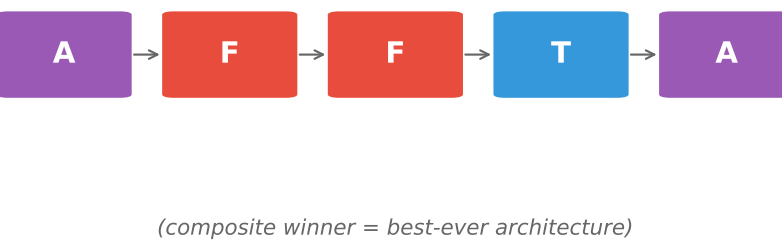
16 labs × 20 iter × 40 epochs × 256 train

Composite fitness: 0.6246

hidden ch.\,= 96, modes = 16, act = relu



Discovered architecture
composite winner - 5 blocks

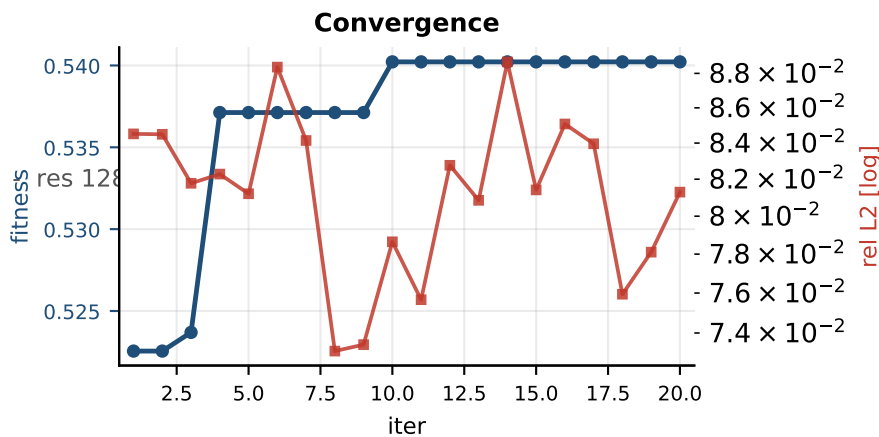


Burgers (1D)

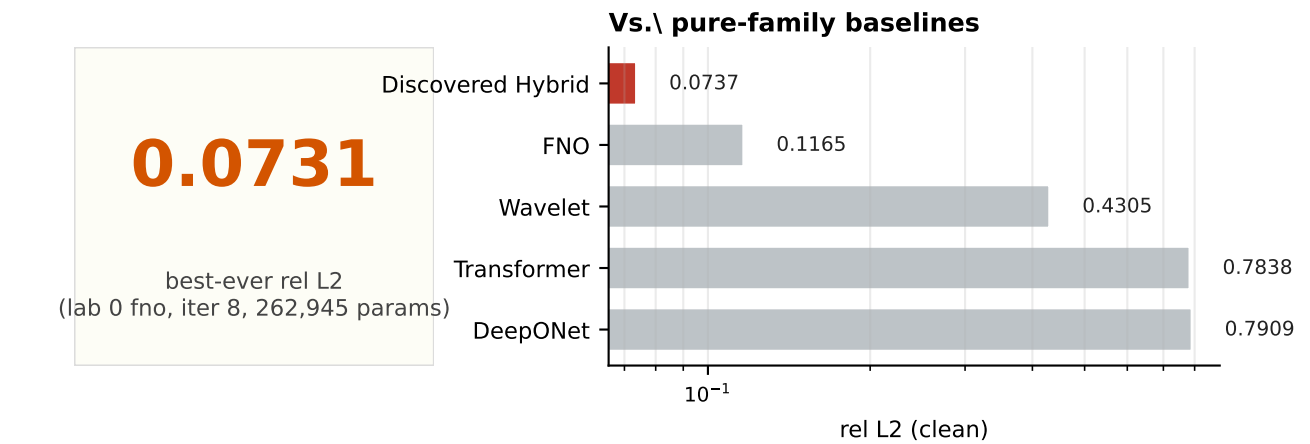
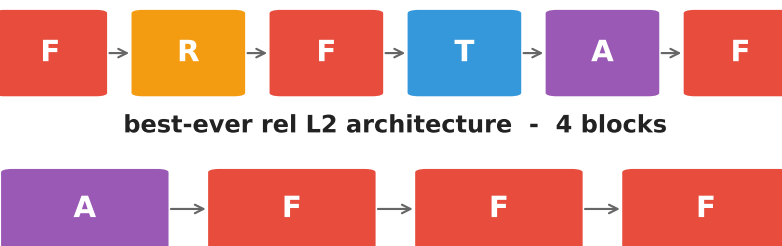
16 labs × 20 iter × 40 epochs × 256 train

Composite fitness: 0.5402

hidden ch.\,= 64, modes = 16, act = relu



Discovered architecture
composite winner - 6 blocks

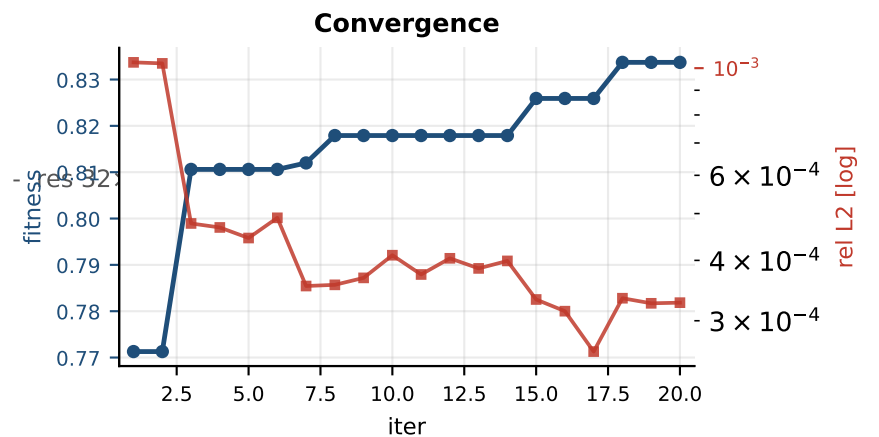


Navier--Stokes (2D)

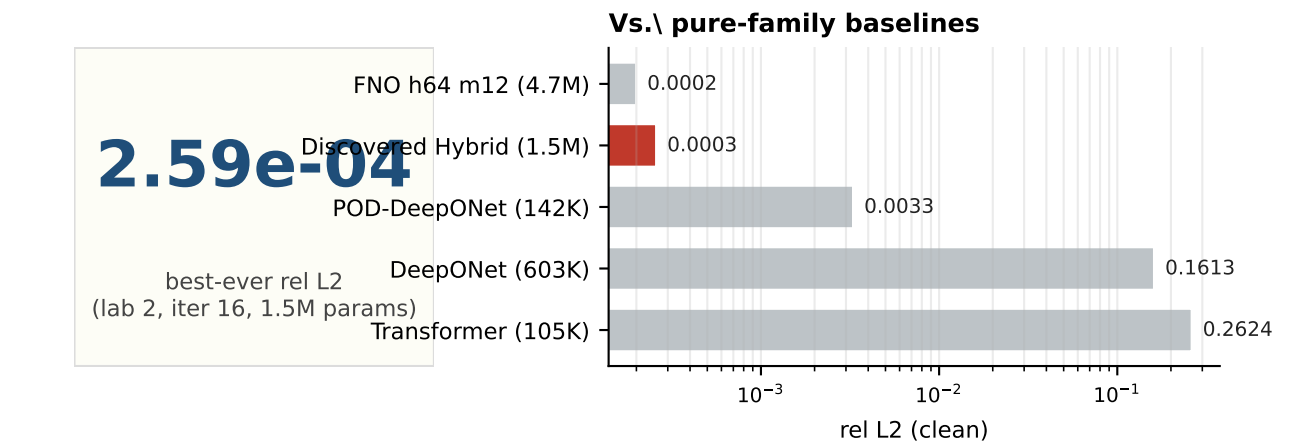
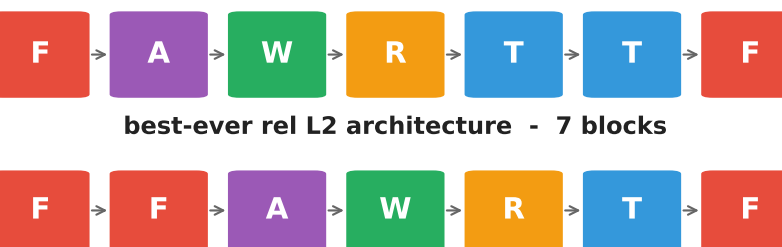
16 labs × 20 iter × 50 epochs × 512 train

Composite fitness: 0.8337

hidden ch.\,= 64, modes = 10, act = gelu



Discovered architecture
composite winner - 7 blocks



Fourier (FNO)

Attention (Transformer / GNOT)

Wavelet (MWT)

Residual Conv (UNet-style)

Branch-Trunk (DeepONet)

Block library used by every lab; the 5-letter pipeline above each PDE row is the genome that the swarm converged on.